



NYC Citi Bike Rider Behavior and Data Insights

Understanding user patterns to optimize station usage.

Citi Bike Ridership Analysis (Winter, Spring, Summer 2025)

Dataset Overview

- The analysis uses NYC Citi Bike ride data from Winter, Spring, and Summer 2025.
- Original dataset contained over 11 million ride records.
- Data was cleaned, validated, and combined into a master analytical dataset.
- Created a 10% stratified seasonal sample to maintain ridership patterns and improve Python and Google Colab processing efficiency.



Business Questions



Network Overview

Where does Citi Bike activity occur?

- Station activity
- Geographic patterns
- High-demand locations

Rider Behavior

Who uses Citi Bike and how?

- Member vs Casual riders
- Seasonal trends
- Trip patterns



Bike Usage Patterns

What bicycle types do riders prefer?

- Electric vs Classic
- Rider preferences (am/pm)
- Weekdays vs Weekend

Motivation: Understanding ridership patterns, station demand, and bike preferences can support better network planning, resource allocation, and rider experience.

Project Methodology (Phase I & II)

Phase I – Python Data Cleaning, Validation, and Analysis

• Data Acquisition & Preparation

- Combined row data for February, May, and August 2025 Citi Bike datasets (~11.4M rides)
- Created master dataset: **citibike_clean.csv**
- Tools: **Pandas, NumPy, Matplotlib, Seaborn**

• Data Validation & Cleaning

- Converted started and ended columns to datetime
- Calculated trip duration (duration_min)
- Removed invalid rides:
 - Duration ≤ 0 minutes
 - Duration > 1440 minutes (24 hours)
- Removed rows with missing station or coordinate data

• Statistical Analysis

- Q1 = 5.27 min
- Median = 9.08 min
- Q3 = 15.83 min
- Mean = 13.36 min
- IQR analysis used to investigate outliers



• Feature Engineering

- Created Analytical fields for visualization using Matplotlib and Tableau Public
 - duration_min
 - Hour (of the day)
 - day_of_week
 - month
 - season
 - is_weekend

Goal: Transform Citi Bike trip data into actionable insights on station utilization, rider behavior, and bicycle preferences across New York City.

Project Methodology (Phase I & II)

Phase II – Dataset Preparation & Business Validation

Visualization: Dataset Preparation for Tableau

- Used the cleaned dataset: **citibike_clean.csv**
- Created 10% stratified seasonal sample (~1.14M rides) to reduce processing requirements
- Preserved Winter, Spring, and Summer ridership patterns
- The Stratified dataset Improves analysis and visualization efficiency.

Business Analytics Validation

- Compared Station IDs vs Station Names
- Performed geographic validation
- Calculated station departures and arrivals
- Created Total Activity metric
- Tools: **Pandas, NumPy, Matplotlib, Seaborn, and**
- **Public Tableau:** *Interactive dashboards* and advanced visualizations

Output Datasets

Dataset 1

citibike_tableau_clean.csv (~1.14M rides)

- Ride-level analytical dataset
- Includes derived fields (duration, hour, day of week, month, season, weekend)
- Supports rider behavior and usage analysis

Dataset 2

citibike_station_activity_by_name.csv

- Station-level analytical dataset
- Aggregates departures, arrivals, and total activity
- Supports station popularity and network analysis

Project Workflow

Phase I



CollectedRaw Citi Bike Data (Feb, May, Aug 2025)

- 🖌️ Data Cleaning & Validation
- 📊 Statistical Analysis (Q1, Median, Q3, IQR, Quartiles Analysis)
- ⚙️ Feature Engineering (duration_min, season, hour, day_of_the_week, is_weekend)
- 💾 citibike_clean.csv (11.4 Million rides)

Source Notebook citibike_seasonal_dataanalytics.ipynb

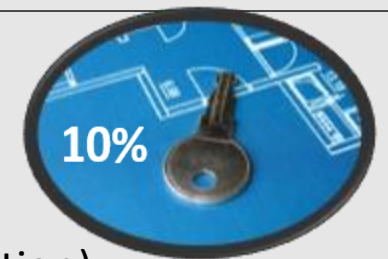
Citi Bike Trip Data Archive: <https://s3.amazonaws.com/tripdata/index.html>

Phase II – Sampling, Validation and Visualization



citibike_clean.csv

- 🖌️ Created 10% Stratified Seasonal Sample dataset (Key enabler)
- ✅ Performed Business Validation (Station Analysis, Geography and Sampling Validation)
- 📁 Created Tableau Datasets (citibike_tableau_clean.csv, citibike_station_activity_by_name.csv)
- 📈 Dashboards & Insights



Source Notebook Citibike_seasonal_DataAnalytics_python_sampled_phase2.ipynb

Network Overview

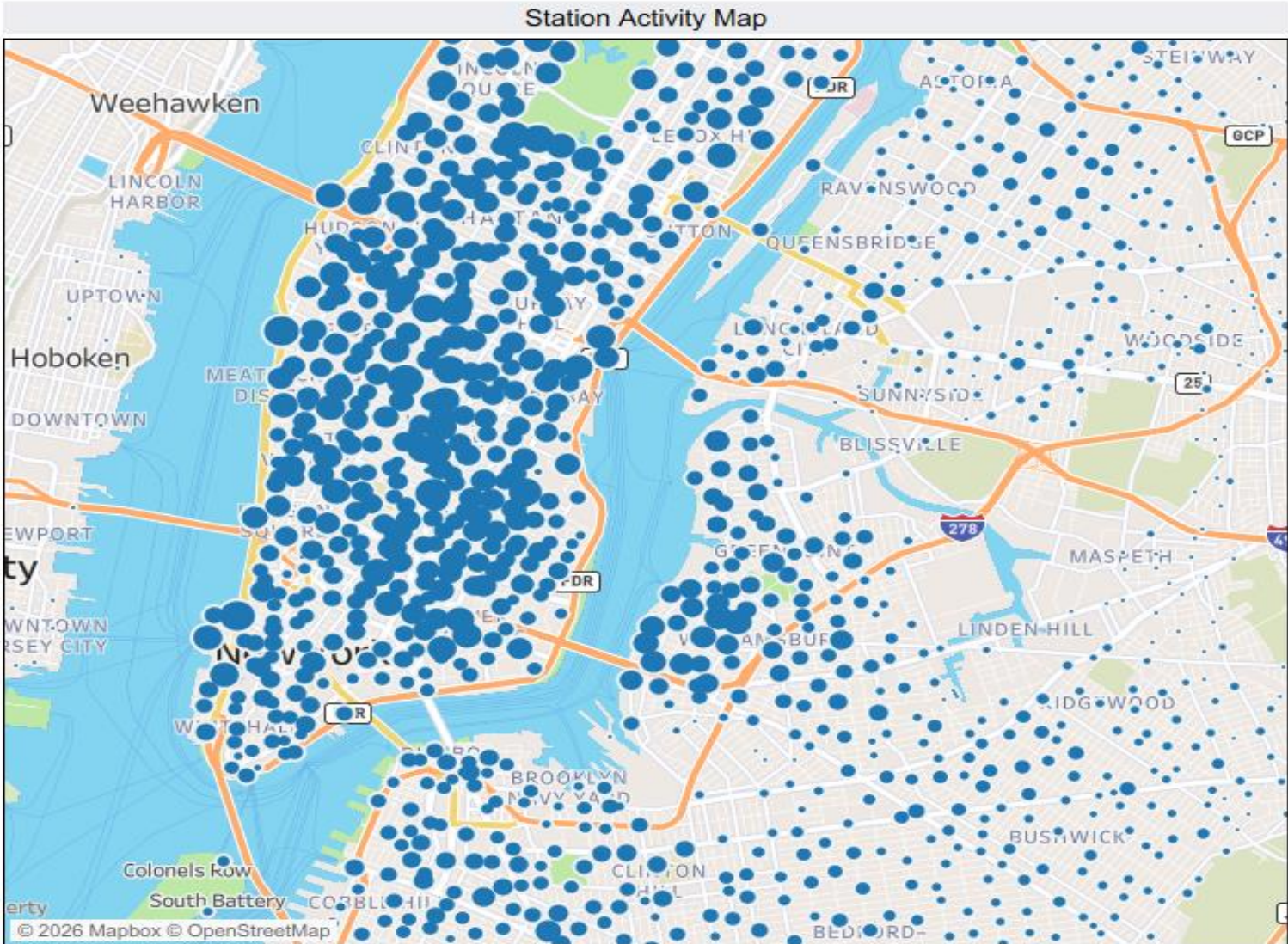
Where does Citi Bike activity occur across New York City's bike-sharing network, and which stations generate the highest and lowest levels of demand?

Top 20 Stations by Total Activity

Station Name		
W 21 St & 6 Ave		7,994
Pier 61 at Chelsea Piers		7,952
Lafayette St & E 8 St		7,256
West St & Chambers St		7,245
W 31 St & 7 Ave		6,539
9 Ave & W 33 St		6,488
Broadway & E 14 St		6,457
7 Ave & Central Park South		6,404
University Pl & E 14 St		6,374
11 Ave & W 41 St		6,353
E 17 St & Broadway		6,261
Broadway & W 58 St		6,177
Cleveland Pl & Spring St		6,161

Bottom 20 Stations by Total Activity

Station Name		
Astor Place		1
Bloomfield St & 15 St		1
Columbus Park - Clinton St & ..		1
Dey St		1
Dixon Mills		1
Exchange Pl		1
Grand St & 14 St		1
Hamilton Park		1
JC Medical Center		1
Jersey & 3rd		1
Manila & 1st		1
River St & 1 St		1
Stevens - River Ter & 6 St		1



Citi Bike activity is unevenly distributed across the network. While ridership is distributed throughout the network, several stations generate substantially more activity than the system average. The station activity map highlights demand concentration across the NYC service area, while the ranking charts identify both the busiest and least utilized locations.

Rider Behavior Analysis

Who uses Citi Bike, how does ridership change across seasons, and how do riding patterns differ between Members and Casual ride..



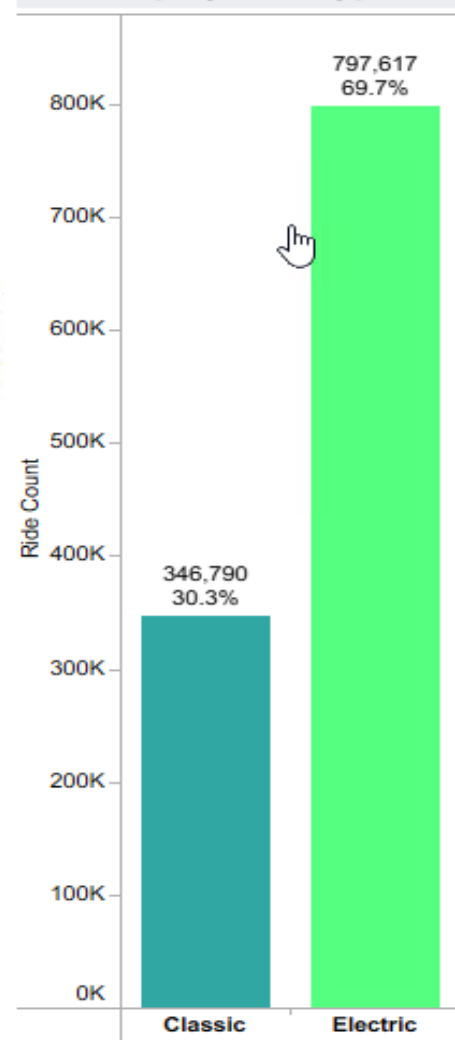
Key Insights

- Members account for 81.1% of all Citi Bike rides.
- Summer is the busiest season.
- Member ridership peaks during morning /evening commute..
- Casual riders take longer trips on average, suggesting different usage patterns than Members.
- Saturday is the busiest day of the week

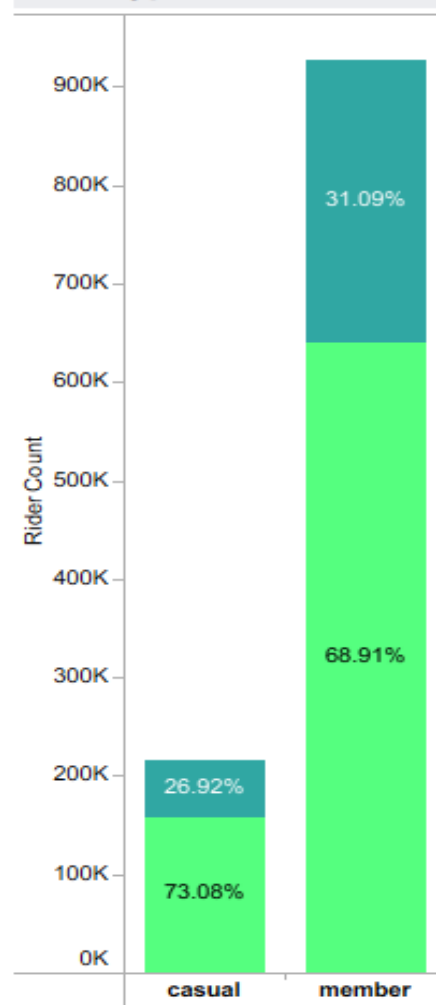
Bike Usage Patterns

What bicycle types do riders prefer across rider groups, seasons, and usage patterns?

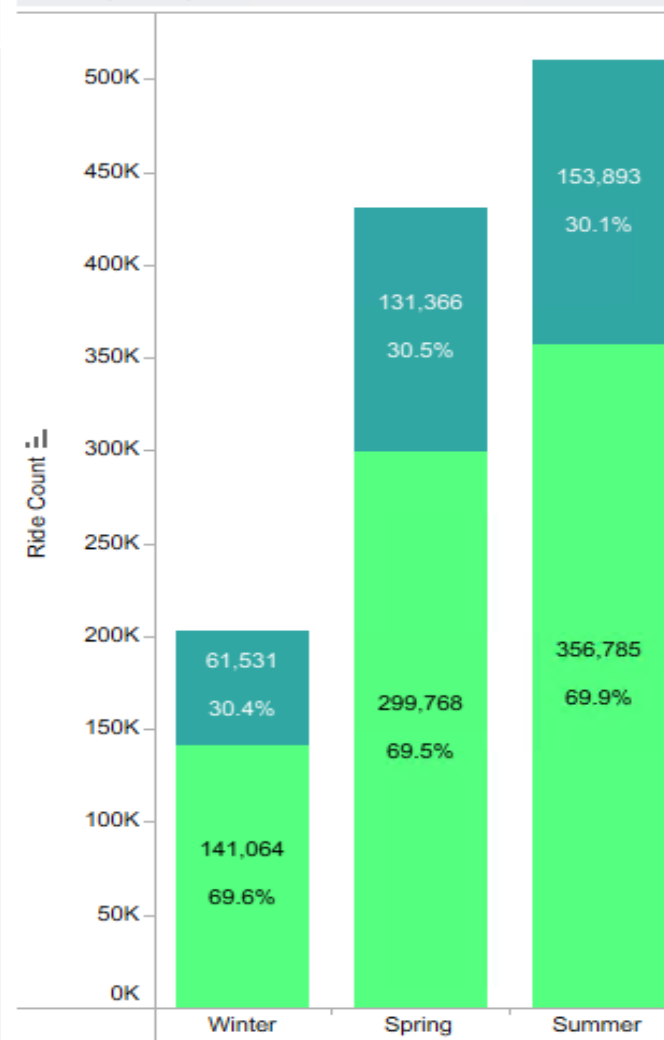
Ridership by Bike Type



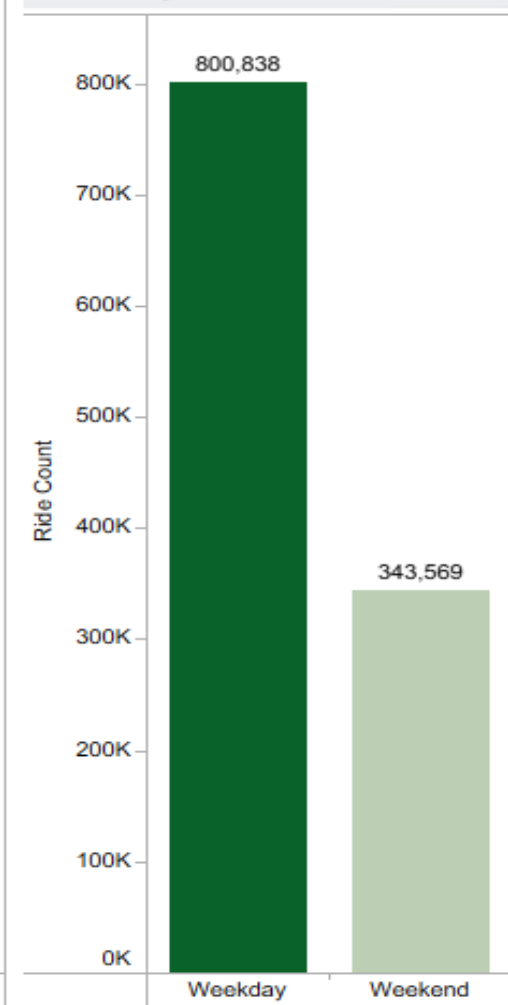
Bike Type Composition by Rider Type



Bike Type by Season



Weekday vs Weekend Ridership



Key Insights

- Electric bicycles account for approximately 70% of all Citi Bike rides.
- Both Members and Casual riders prefer electric bikes over classic bikes.
- Electric bicycles have become the dominant mode of Citi Bike usage across rider segments
- Casual riders show an even stronger preference for electric bicycles.
- Bike-type preferences remain relatively consistent between weekdays and weekends...

Business Recommendations

- Expand electric-bike availability.
- Focus capacity planning on high-demand locations.
- Prepare for increased summer casual ridership.
- Optimize bike availability during commuter peaks.
- Promote memberships to frequent casual riders.

Final Conclusions

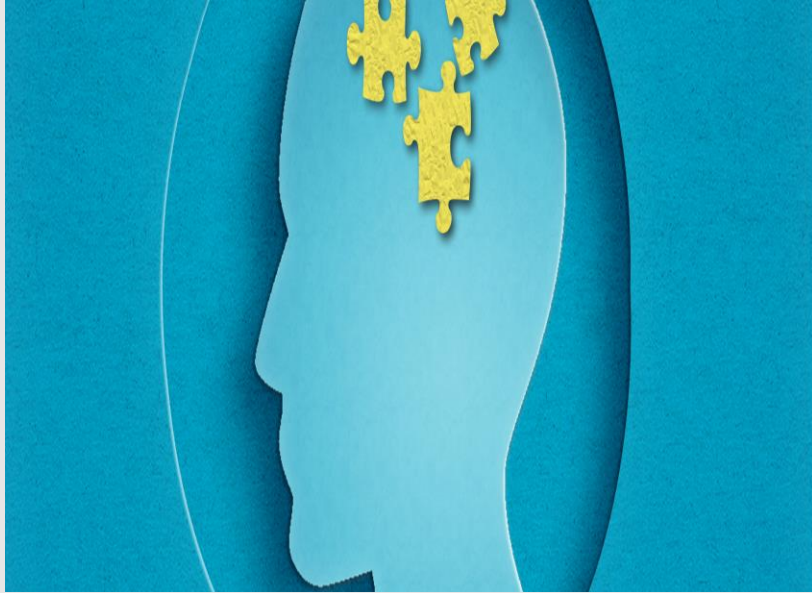
KEY FINDINGS

- **Members** account for **81.1%** of all Citi Bike rides.
- **Summer** is the **busiest** riding season.
- Casual rider participation more than doubles during Summer.
- **Member** ridership exhibits strong **commuter-oriented** behavior.
- Weekend ridership remains strong, indicating both transportation and recreational use.
- **Electric bicycles** account for approximately **70%** of all rides.
- Station activity is unevenly distributed across the network.
- The station rankings and activity map show that some locations generate substantially higher activity than others.

FINAL TAKEAWAY

Citi Bike primarily operates as a transportation network supporting daily mobility throughout New York City. **Member riders drive the majority of system usage**, suggesting a strong commuting and routine travel component. However, consistent weekend demand, increased **Summer** ridership, and higher participation among Casual riders indicate that the system also serves recreational, leisure, and tourism-related activities. Together, these **patterns demonstrate that Citi Bike fulfills both transportation and recreational roles**. Regardless of Membership type; **electrical bikes** have become the **dominant mode of Citi Bike usage**.





Thank you

Any Questions ? / Comments

